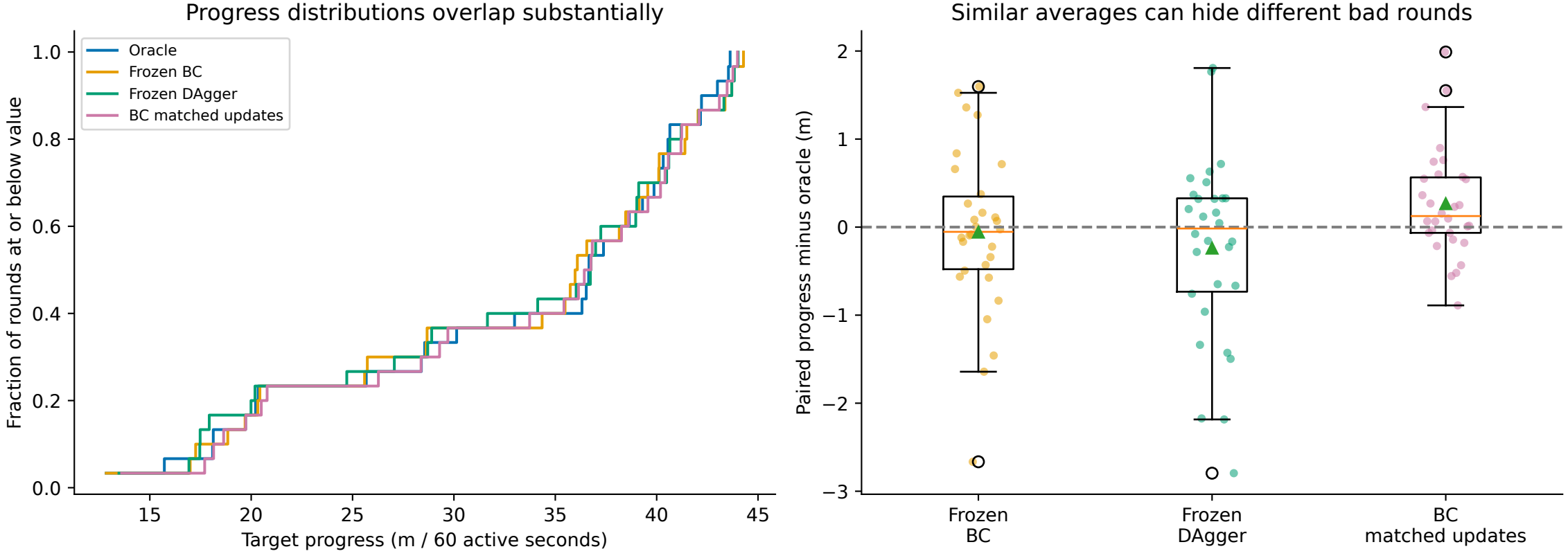
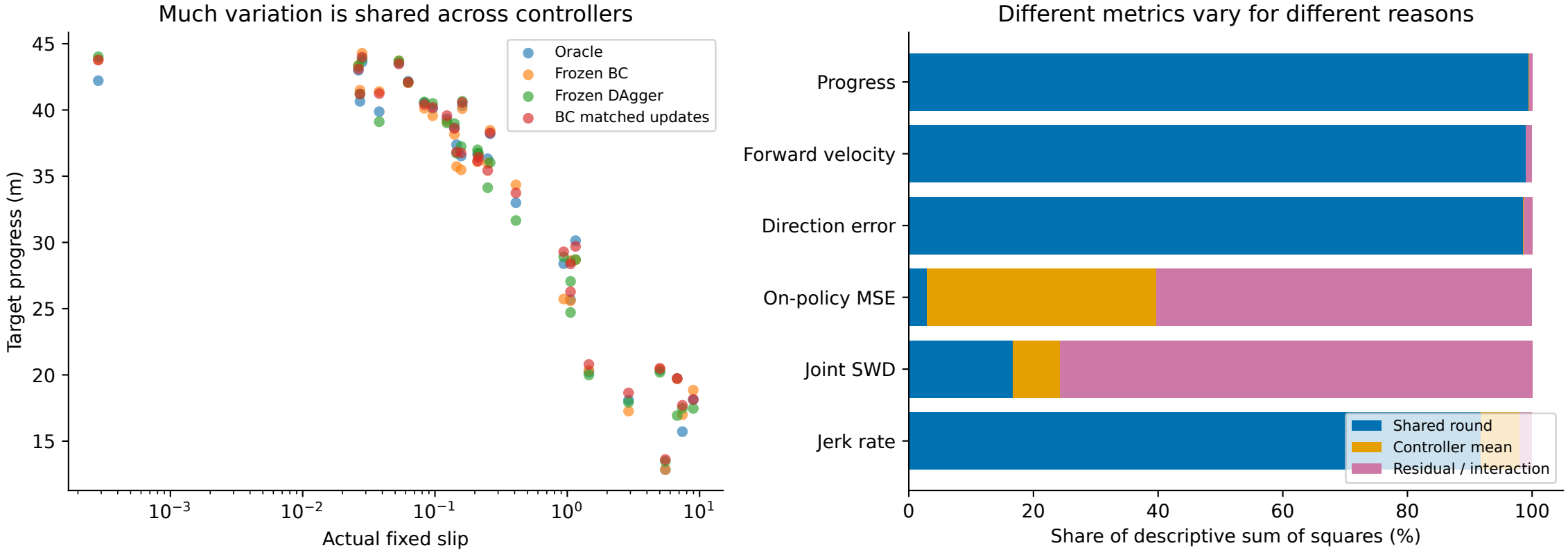
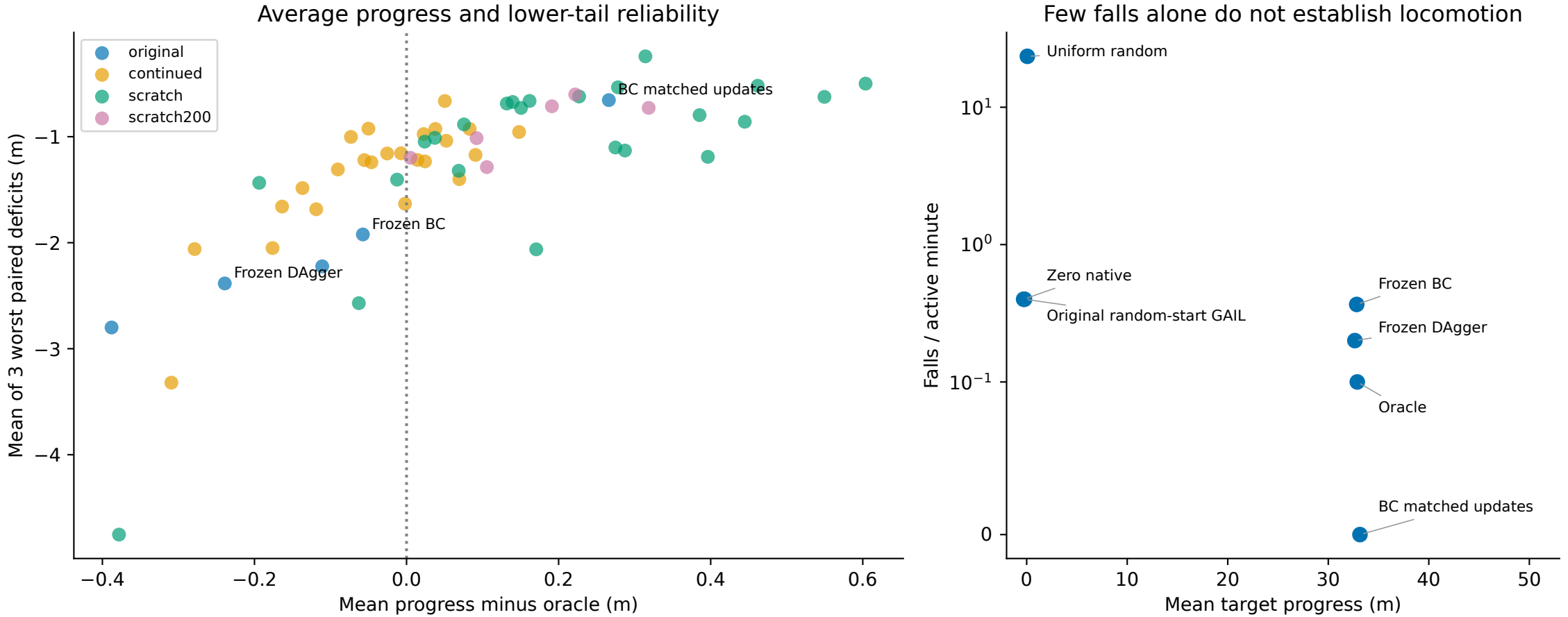


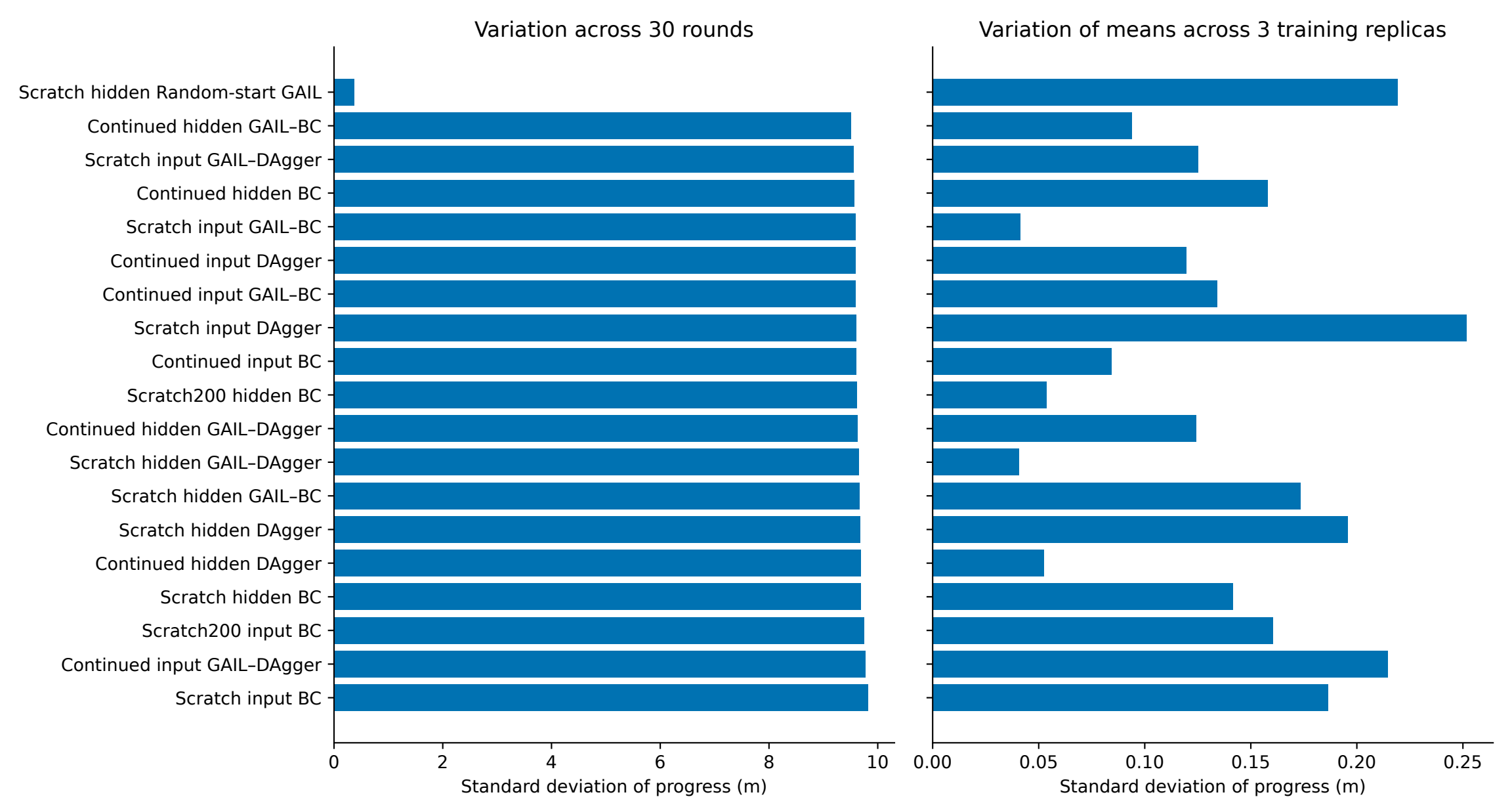
30 shared surface/task rounds per model; whiskers = 1.5 IQR; dots = rounds; triangle = mean. No training-seed CI.



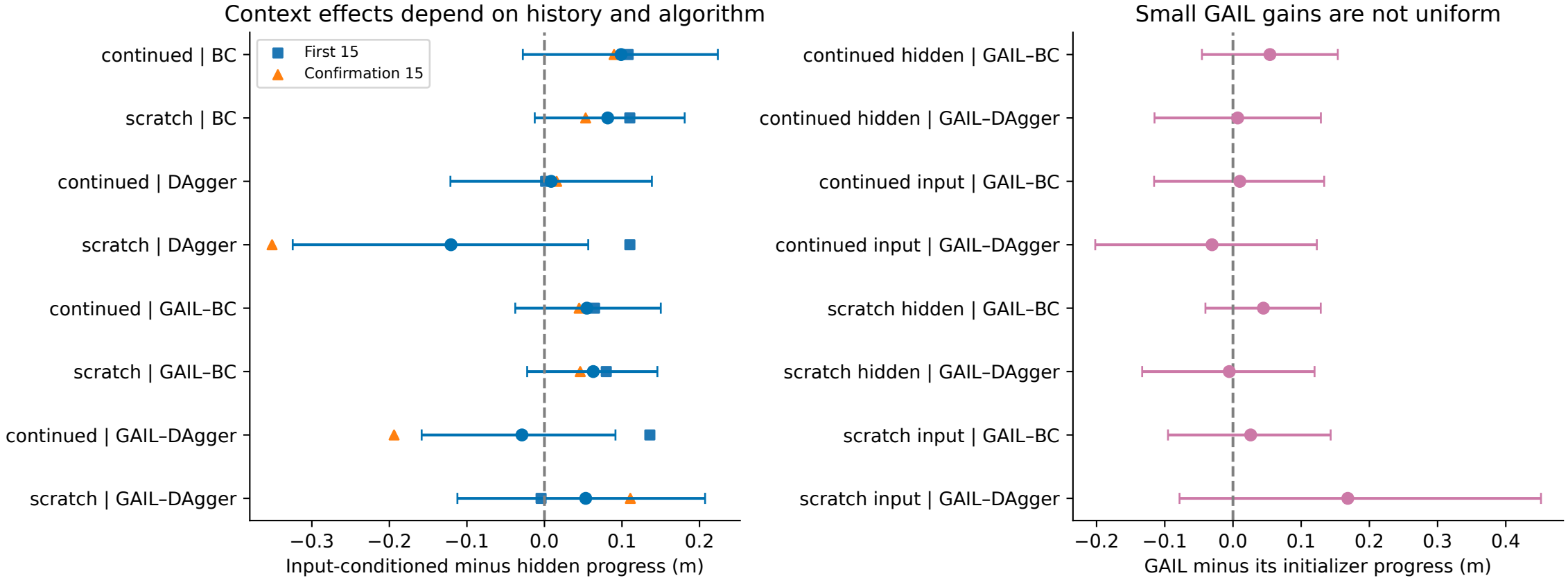
Balanced panel of walking controllers; round includes surface AND task. Residual includes simulator variation; this is not causal ANOVA.



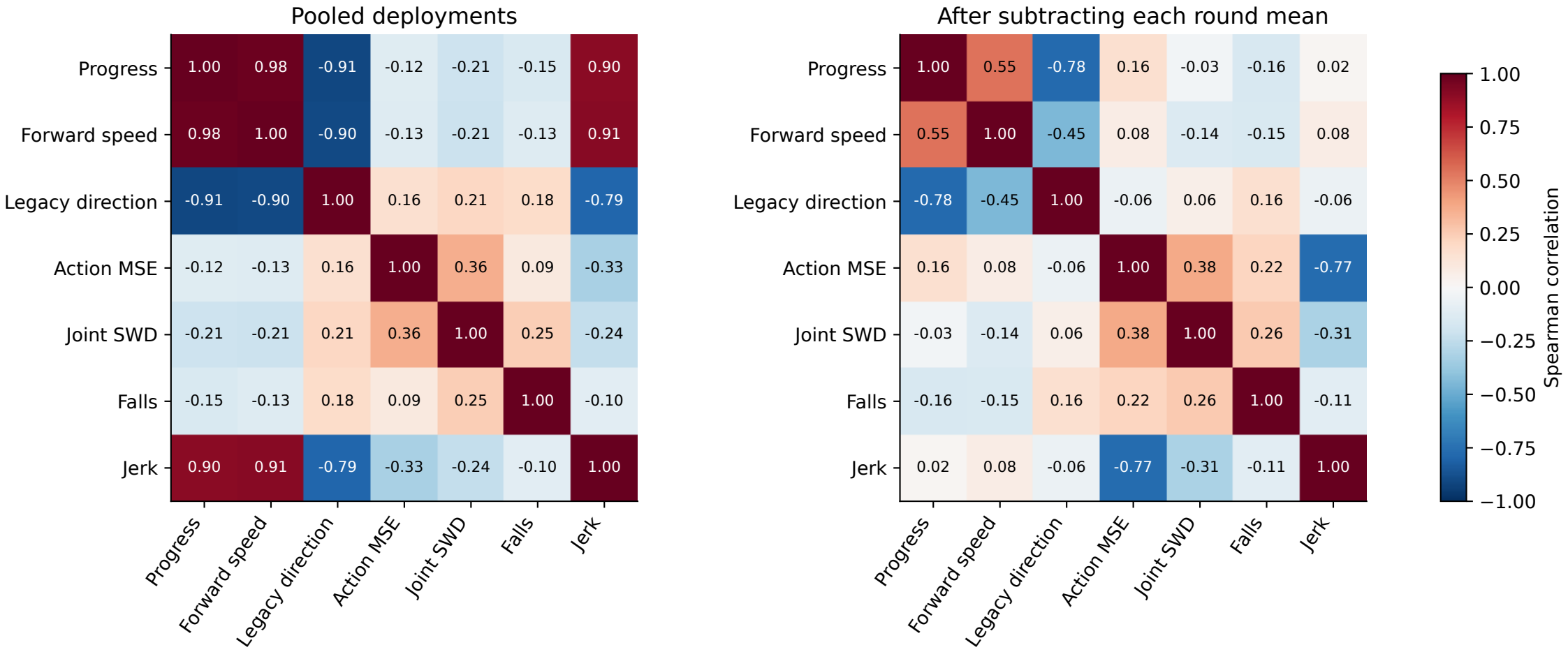
Left: 30 rounds/checkpoint; tails are descriptive order statistics. Right: same rounds, symmetric-log fall axis (linear below 0.1).



These standard deviations describe different quantities. Continued replicas share an original parent; n=3 gives a weak estimate of training variability.

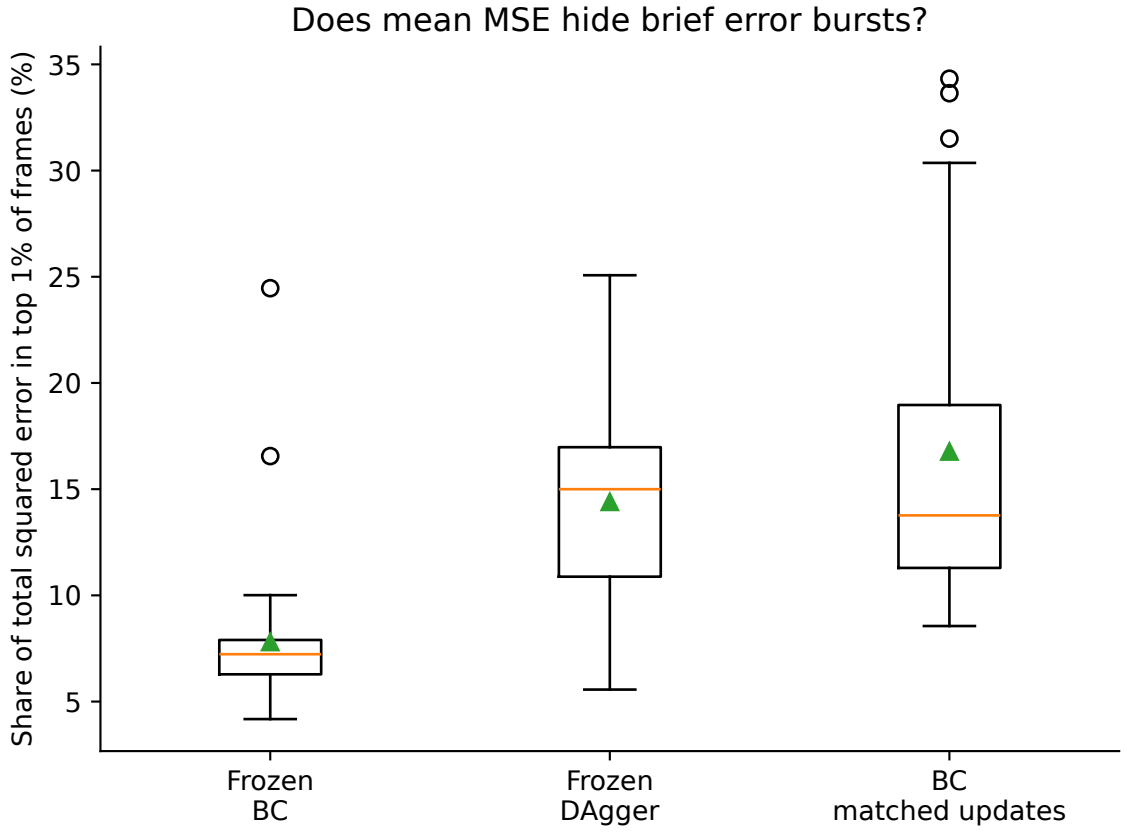


95% stratified round-bootstrap intervals conditional on observed checkpoints; exploratory, no multiplicity correction or population-of-training-seeds claim.

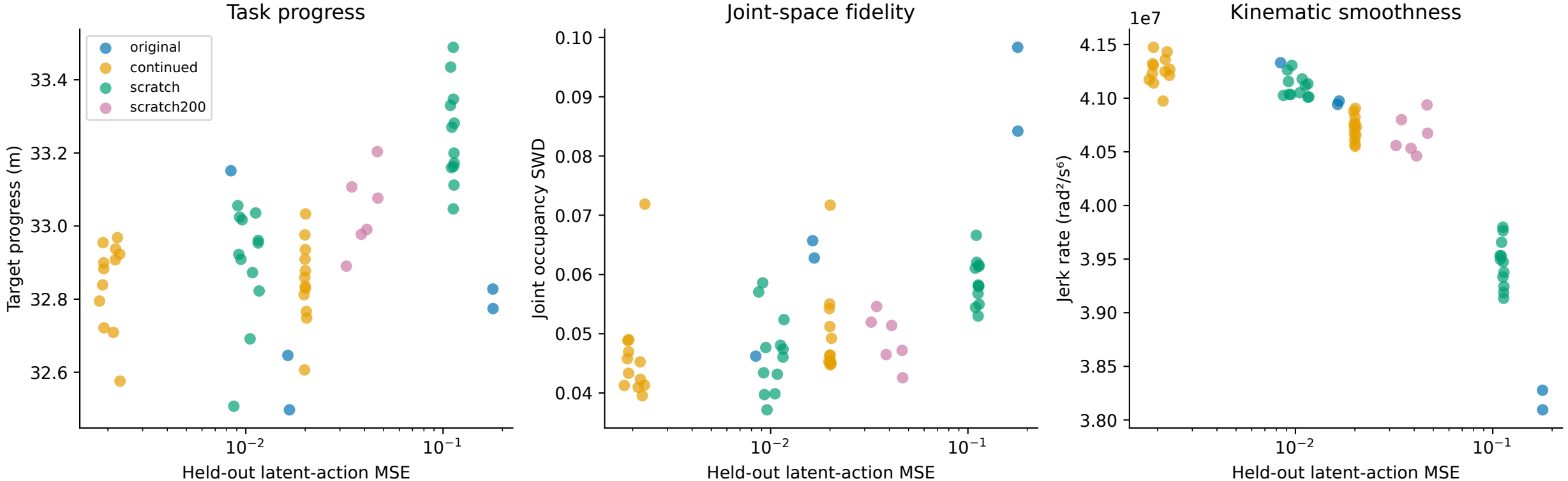


Walking policies only; dependent deployments, descriptive correlations. Round centering removes shared round levels, not all confounding.





Each histogram entry and box-plot observation is a deployment, not an independent time step. Top 1% = 30 of 3,000 frames.



One point per checkpoint, 30-round means. Shared held-out test pool; architectures and training histories are mixed, so trends are not causal.

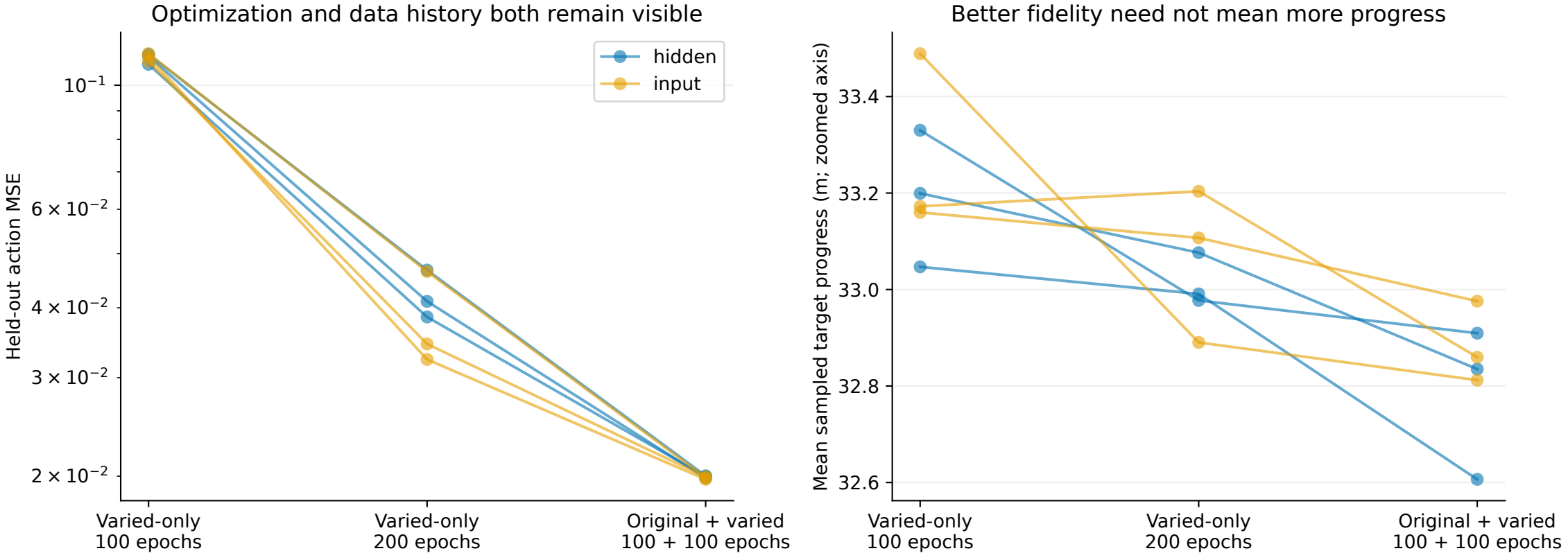
Gain over matched random-network readouts

Gain over raw-observation readouts

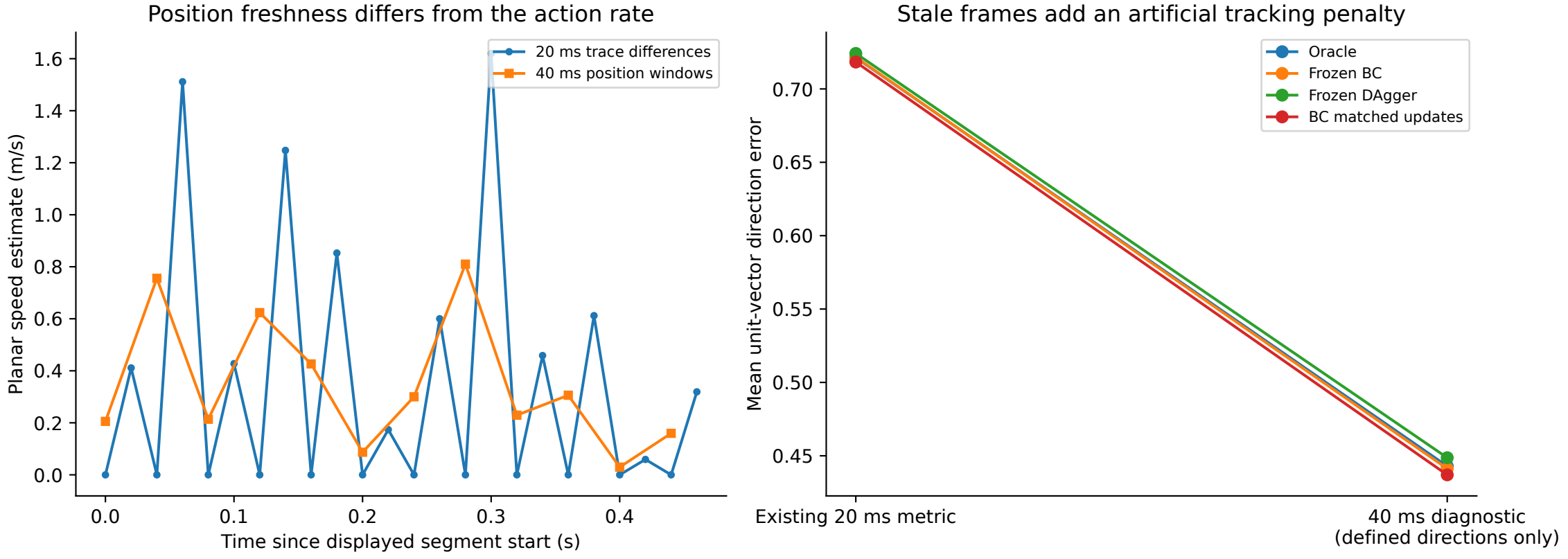
	Gain over matched random-network readouts				Gain over raw-observation readouts			
	bc_h1	bc_h2	DAgger_h1	DAgger_h2	bc_h1	bc_h2	DAgger_h1	DAgger_h2
command forward displacement m	+0.02	+0.06	+0.03	+0.07	-0.01	-0.02	-0.00	-0.01
command lateral displacement m	+0.05	+0.12	+0.05	+0.13	-0.01	-0.02	-0.00	-0.02
command orientation deg	+0.05	+0.14	+0.05	+0.14	-0.00	-0.01	-0.00	-0.00
future 200ms direction error	+0.04	+0.07	+0.04	+0.06	+0.04	+0.03	+0.05	+0.01
future 200ms forward velocity mps	+0.02	+0.03	+0.02	+0.04	+0.03	+0.02	+0.04	+0.03
future 200ms left force contact gt10N	+0.01	+0.03	+0.02	+0.03	+0.02	+0.02	+0.02	+0.02
future 200ms pitch deg	+0.03	+0.06	+0.03	+0.06	+0.04	+0.04	+0.03	+0.04
future 200ms right force contact gt10N	+0.01	+0.03	+0.01	+0.02	+0.03	+0.02	+0.02	+0.02
future 200ms roll deg	+0.05	+0.11	+0.06	+0.11	+0.02	+0.03	+0.04	+0.02
generator phase cos	+0.00	+0.01	+0.00	+0.01	+0.00	-0.00	+0.00	-0.00
generator phase sin	+0.02	+0.03	+0.02	+0.03	-0.01	-0.01	-0.01	-0.01
gyro x degps	+0.04	+0.10	+0.05	+0.11	-0.02	-0.08	-0.01	-0.06
gyro y degps	+0.04	+0.11	+0.04	+0.08	-0.02	-0.07	-0.02	-0.10
gyro z degps	+0.01	+0.02	+0.01	+0.01	-0.01	-0.04	-0.01	-0.06
head height m	+0.07	+0.06	+0.15	+0.12	-0.37	-0.44	-0.29	-0.38
head vertical velocity mps	+0.20	+0.07	+0.38	+0.23	-0.31	-0.53	-0.13	-0.37
left ankle hip x m	+0.02	+0.03	+0.02	+0.04	-0.01	-0.02	-0.01	-0.01
left ankle hip y m	+0.03	+0.04	+0.05	+0.07	-0.07	-0.12	-0.05	-0.09
left ankle hip z m	+0.01	+0.03	+0.02	+0.03	-0.04	-0.05	-0.03	-0.05
left force contact gt10N	+0.03	+0.04	+0.04	+0.05	+0.00	-0.02	+0.01	-0.01
left force norm N	+0.04	+0.08	+0.05	+0.12	-0.02	-0.08	-0.01	-0.04
phase cos times pitch deg	+0.04	+0.03	+0.06	+0.05	+0.07	+0.03	+0.09	+0.05
phase sin times roll deg	+0.23	+0.26	+0.26	+0.24	+0.32	+0.29	+0.35	+0.27
right ankle hip x m	+0.02	+0.02	+0.02	+0.03	-0.01	-0.03	-0.01	-0.02
right ankle hip y m	+0.06	+0.06	+0.08	+0.10	-0.09	-0.17	-0.07	-0.13
right ankle hip z m	+0.01	+0.02	+0.02	+0.03	-0.02	-0.03	-0.02	-0.02
right force contact gt10N	+0.02	+0.04	+0.03	+0.04	-0.00	-0.02	+0.00	-0.01
right force norm N	+0.05	+0.10	+0.06	+0.13	-0.03	-0.08	-0.01	-0.06
torso pitch deg	+0.10	+0.14	+0.13	+0.18	-0.05	-0.12	-0.02	-0.08
torso roll deg	+0.16	+0.29	+0.17	+0.30	-0.04	-0.10	-0.02	-0.08

Difference in held-out R^2: 0.4, 0.2, 0.0, −0.2, −0.4

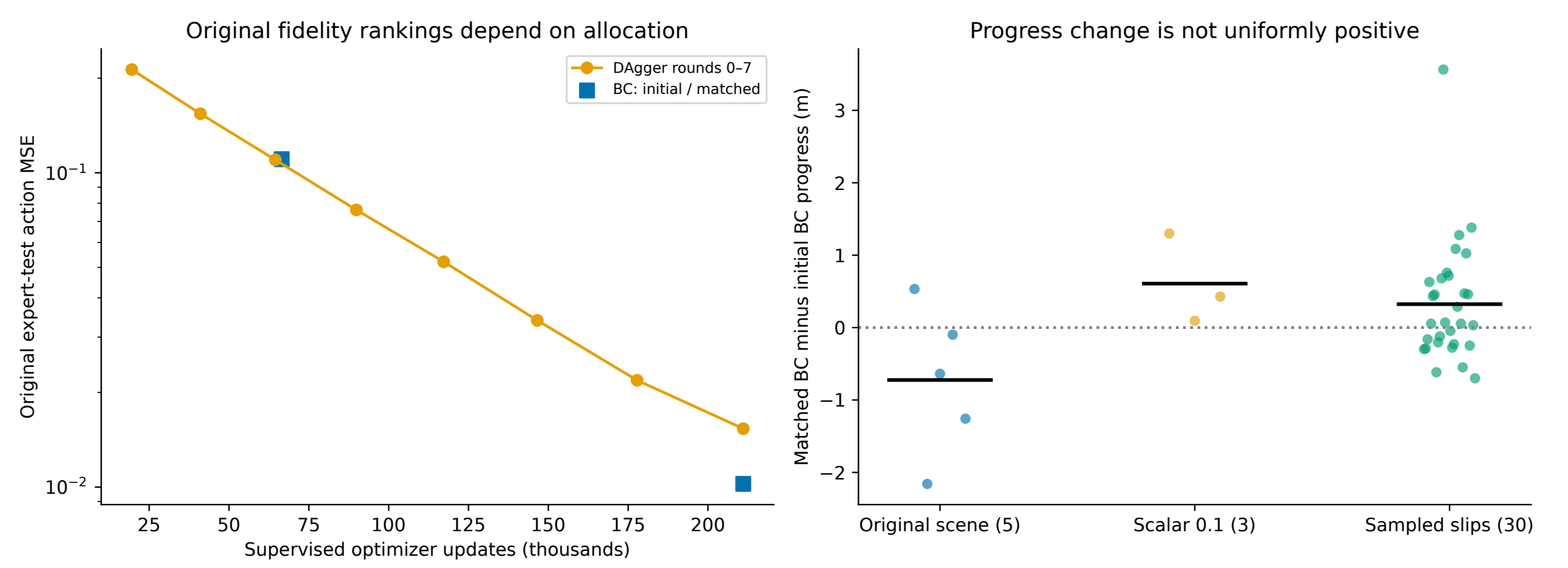
All 30 pre-existing probe targets; same oracle-state corpus. Colors saturate at ±0.5; printed values are exact to 2 dp. No new probes fitted.



Lines pair seed labels, not continuous training for the continued arm; three replicas per architecture. 200-epoch fits have not established convergence.



Left: one sampled oracle segment. Right: 30 sampled rounds/model. Diagnostic excludes reset/target-change windows; historical metrics unchanged.



Left: nearby initial BC/DAgger points are 66,500 / 64,500 updates, not equal. Right: paired rounds; black lines are means; unequal suite sizes.